Merging all the datasets

We know we need to apply offsets bc GRA is around 0 while others 250 mGal

Dataset	Number of stations	Min [mGal]	Max [mGal]	Amplitude [mGal]
AP	159	216.630	282.667	66.037
CSIC	95	219.833	266.709	46.877
IGN	541	205.040	281.847	76.807
GRA	377	-27.534	60.972	88.506

Script that highlights the station close to 100 m and after check if correct.

BEST SOLUTION :

don't put the free-air gradient bc values already at datum

OFFSETS

Offsets by Least square solution (not taking GRA bc shifted and not same corrections) at paired station close at 0.001 min approx 100m distance.
IGN CSIC and AP had the same corrections so just need a plane horizontal offset to align the values.

For GRA, random 955 offset to raise values but later apply a better fitting model bc not a plane distorsion as the GRA file is complete Bouguer Anomaly.

```
==================================================
🎯 NOUVEAUX OFFSETS PURS (SANS INTERFÉRENCE DE GRA) :
==================================================
-> Offset AP : -0.000 mGal
-> Offset CSIC : +0.451 mGal
-> Offset IGN : +0.211 mGal
-> Offset GRA : +226.350 mGal
==================================================
```

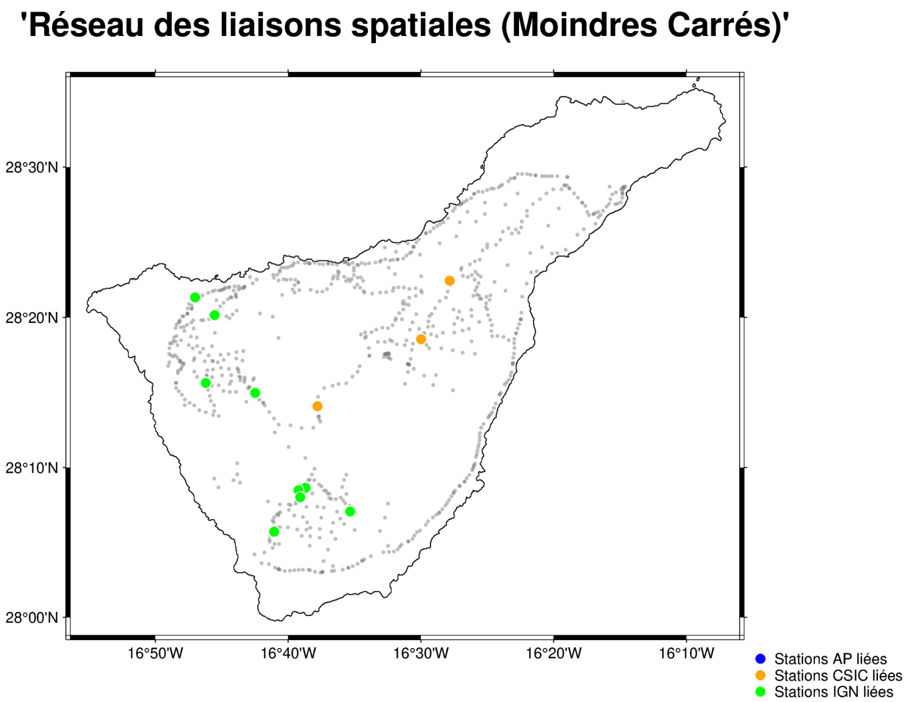
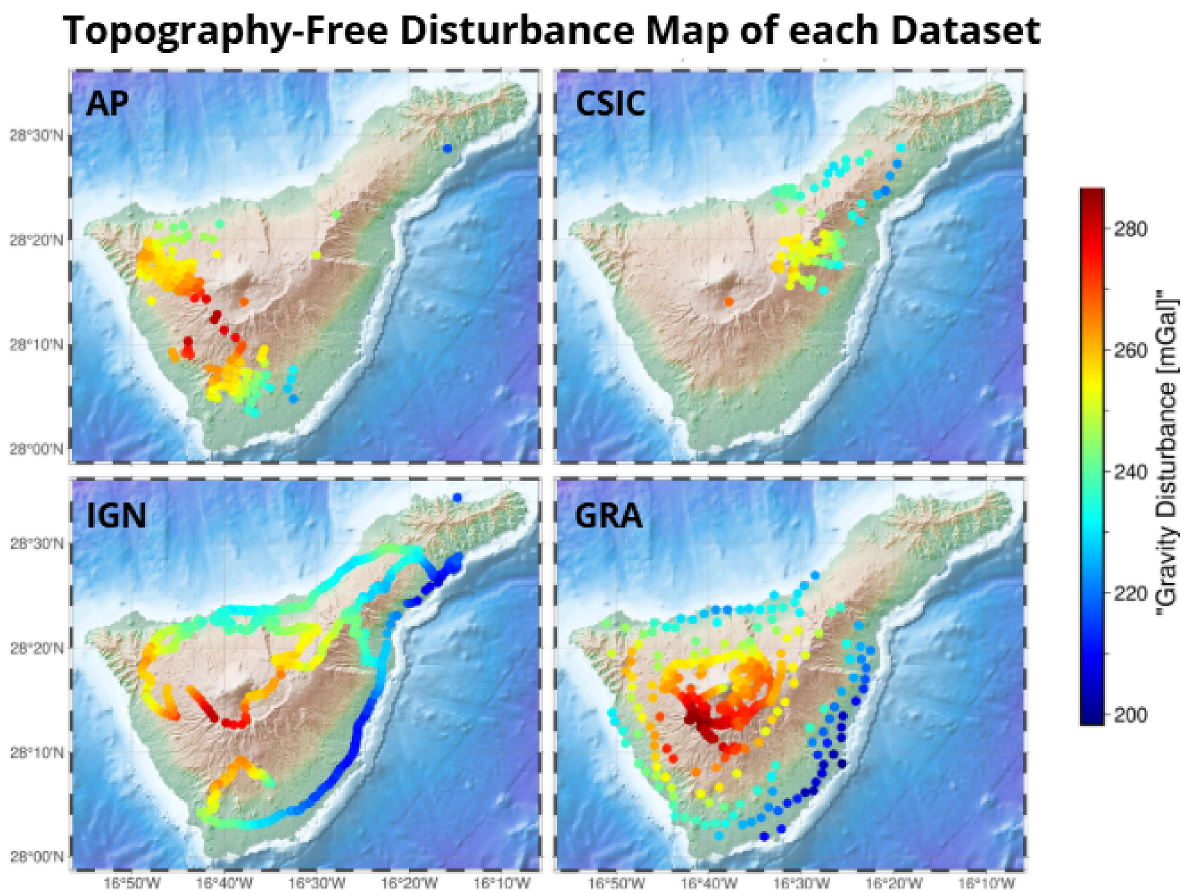


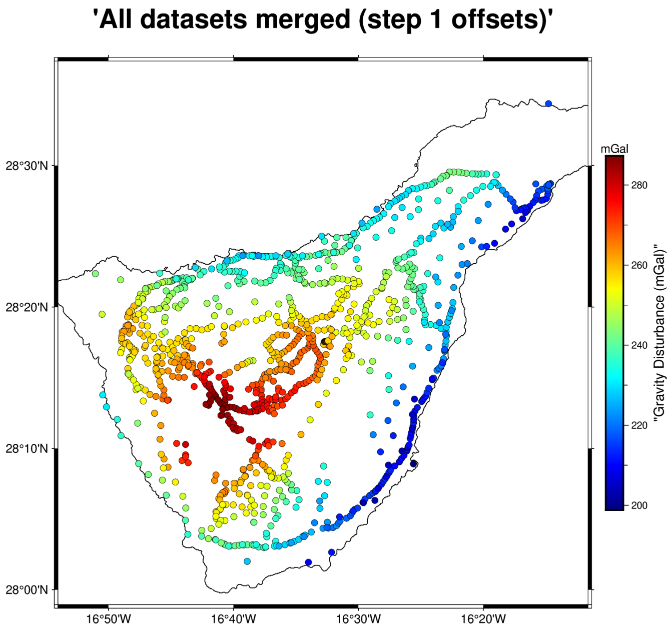
Figure showing the position of the stations that were 100m close.

Check to see if offsets worked : Yes but not GRA which is still far (from CSIC and IGN)

📋 MATRICE DES MISFITS FINALE (POST-OPTIMISATION) :

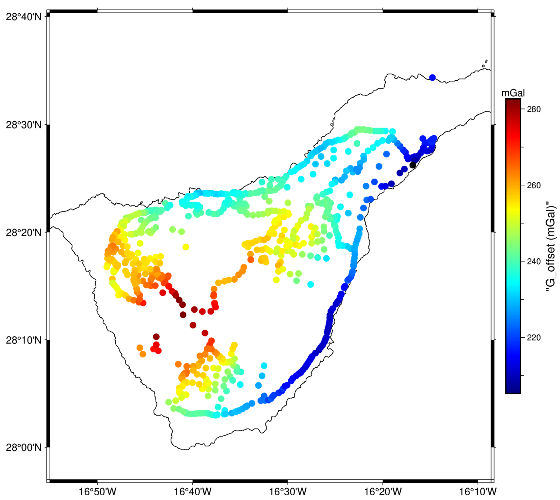
Base D1	Comparée à D2	Nb paires (<100m)	Moyenne Écart (mGal)	Écart-Type (mGal)
IGN	AP	10	-1.136868e-14	0.309175
IGN	GRA	33	1.402141e+00	5.632796
CSIC	AP	3	0.000000e+00	0.449661
CSIC	GRA	3	-5.145092e+00	0.455090
AP	IGN	10	1.136868e-14	0.309175
AP	CSIC	3	0.000000e+00	0.449661
AP	GRA	9	-3.424743e-01	2.987827
GRA	IGN	33	-1.402141e+00	5.632796
GRA	CSIC	3	5.145092e+00	0.455090
GRA	AP	9	3.424743e-01	2.987827


Topography-Free Disturbance Map of each Dataset
AP
CSIC
IGN
GRA
28°30'N
28°20'N
28°10'N
28°00'N
16°50'W
16°40'W
16°30'W
16°20'W
16°10'W
280
260
240
220
200
"Gravity Disturbance [mGal]"

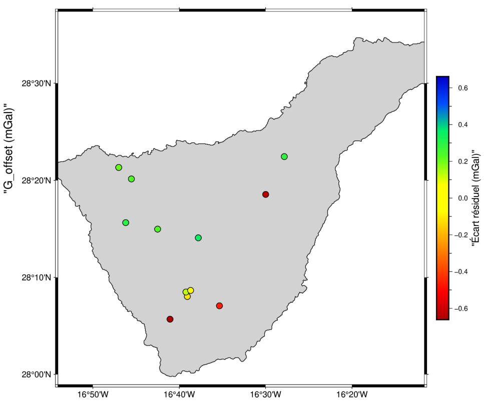

'All datasets merged (step 1 offsets)'
28°30'N
28°20'N
28°10'N
28°00'N
16°50'W
16°40'W
16°30'W
16°20'W
mGal
280
260
240
220
200
"Gravity Disturbance (mGal)"

REFERENCE

'Merged Datasets (IGN CSIC AP)'



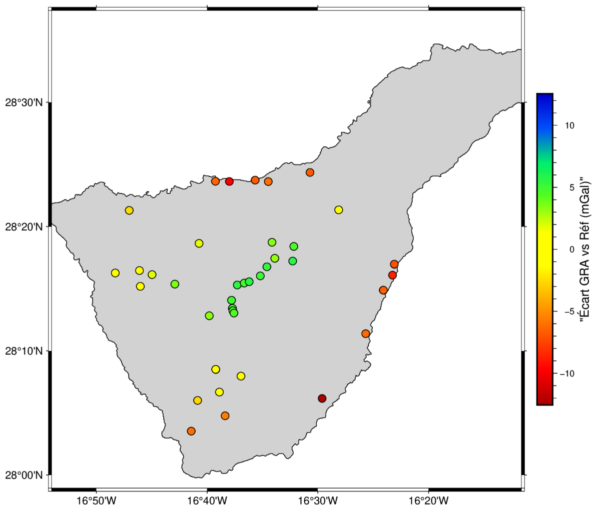
'Misfits résiduels : Bloc de Référence (mGal)'



Merged no gra
These 3 datasets IGN CSIC AP are now almost perfectly merged (>0.5 mGal residual for stations 100m close). They serve as base or reference to next correct GRA.

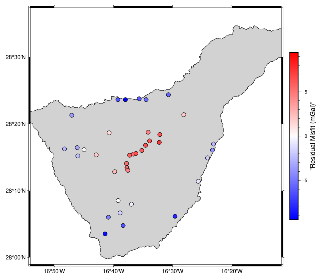
GRA MISFIT :

'Anomalies de misfit locales avec le catalogue GRA (mGal)'



Here we can see a trend on the misfit so we will try to model that trend to correct the dataset.
Linear and bilinear trends don't fit well, still std dev of 5 mGal and until -10 mGal residual.
Model too rigid.

'Residual Misfits Post-Bilinear Correction (mGal)'



So did a second degree quadratic fit, 45 stations close to 100m.

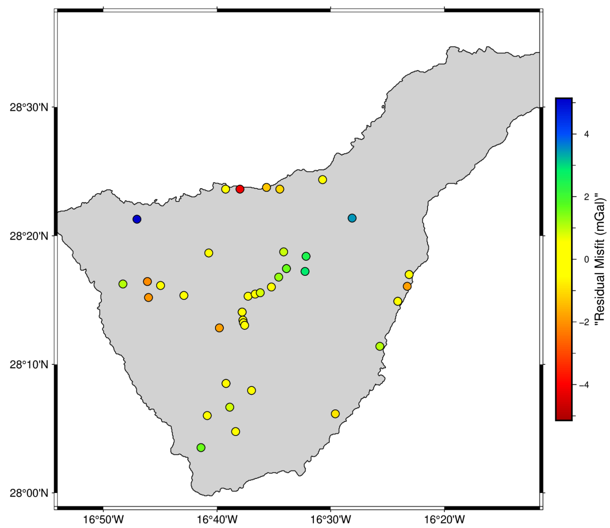
Give this :

📊 VERIFICATION STATISTIQUE DU MODÈLE QUADRATIQUE :

count	4.500000e+01
mean	-2.918268e-09
std	1.588534e+00
min	-4.216888e+00
25%	-6.590336e-01
50%	-3.088813e-01
75%	8.348412e-01
max	5.213000e+00

Std dev at 1.6 so quite good!

'Residual Misfits Post-Quadratic Correction (mGal)'



After quadratic fit of GRA :

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IGN	AP	10	-1.136868e-14	0.309175
IGN	GRA	33	2.201327e-01	1.354786
CSIC	AP	3	9.473903e-15	0.449661
CSIC	GRA	3	-1.793878e+00	1.310487
AP	IGN	10	1.136868e-14	0.309175
AP	CSIC	3	-9.473903e-15	0.449661
AP	GRA	9	-2.091938e-01	1.933984
GRA	IGN	33	-2.201327e-01	1.354786
GRA	CSIC	3	1.793878e+00	1.310487

GRA	AP	9	2.091938e-01	1.933984

FINAL MAP

With all datasets

Used the JET colorbar (or turbo, or seis but too much variations maybe, or cool but not enough variations)